FUTURE TRENDS IN MOBILE COMPUTING

It’s important you’re aware of the new technology that’s available and surrounding you. Who knows, maybe you’ll decide to change your strategy based on this information.

Artificial intelligence (AI)

Artificial intelligence has penetrated our mobile world.

There are advancements in mobile AI. You may be familiar with some of these:

- Alexa
- Siri
- Cortana
- Google Assistant

All of these are examples of AI that may even be installed on your mobile devices right now. In addition to these popular forms of AI, mobile apps are now using software such as voice recognition to encourage hands-free use and ultimately optimize the customer experience.

AI software is used to help developers and marketers learn more about the user.

Businesses are trying to get more revenue by using this information to create relevant advertisements that target specific audiences.

Location-based technology

Your smartphones and tablets are tracking your location.

Mobile applications are also tracking your location, with your permission. Each time you download a new app, it requests your permission to use your location.

Each time you download a new app, you’ll get a notification.

You may not even be able to use some apps to their full potential without giving them access to your location. For example, think about a ride-sharing app such as Uber.

They need your exact location to connect you with a driver.

There is an increase in apps requesting your location even if you don’t think it’s required to use the primary function of the app. That’s because most of the apps on your smartphone share your data with third parties. They do this to enhance their marketing campaigns.

If a business knows where a user is, it can send them targeted ads based on the location. An example of this is when an app uses geofencing technology. How it works - Let's say you own a restaurant and have a mobile app. If an app user walks within a few blocks of your location, they'll receive a notification about your lunch special.

Motion and Location Sensing

Most mobile phones have location sensor capabilities which use multiple positioning methods to provide different granularities of location data. Knowing an individual's location to within a few meters is useful for providing highly relevant contextual information and services. Motion sensing apps are used in security, anti-theft, power-saving and games. Location sensing is useful in Geotagging, Games, Vehicle navigation, and fitness apps. Apps exploiting precise indoor location currently use technologies such as Wi-Fi, imaging, ultrasonic beacons, and geomagnetic. In the longer run technologies such as smart lighting will also become important. Precise indoor location sensing, combined with mobile applications, will enable a new generation of extremely personalized services and information.

Augmented reality

Augmented reality takes something that's real and modifies it.
One of the best examples of this is the face filter options on Snapchat. Recently, Instagram implemented this feature as well.
This will help businesses create brand awareness, app downloads, engagement, and revenue.

Syncing wearable technology with mobile devices

Wearable technology has become increasingly popular. The smartphone will become the hub of a personal-area network consisting of wearable gadgets such as on-body healthcare sensors, smart jewelry, smart watches, display devices (like Google Glass) and a variety of sensors embedded in clothes and shoes. These gadgets will communicate with mobile applications to deliver information in new ways. And will enable a wide range of products and services in areas

such as sport, fitness, fashion, hobbies and healthcare. Thus, wearable devices connected with smartphones will influence the next generation of mobile application development strategies.

Take Fitbit as an example. All the movements of a person wearing it can be tracked through an app. Users can check their heart rates and how many miles they walked in a day, among other things.

By syncing with mobile devices, these apps can be used socially as well. People can compare their progress with their friends and make it a competition. As a result, it encourages the usage of the technology and increases engagement.

Mobile devices syncing with homes

Mobile apps are being developed to help improve consumers' experiences within their own homes.

You can find businesses that sync your home air conditioning and heating with an app. That way, you can control temperatures whether you're home or not. Instead of going to a central thermostat in the house, you can reach into your pocket and set everything on your phone.

Home security has been integrated with mobile technology as well. There are apps that have a video camera synced with your doorbell so you can see who is at your front door when the bell rings. Home security cameras on the inside and outside of your home can all be controlled and monitored from mobile devices.

There are even smart refrigerators that connect with mobile devices. This technology gives you the ability to see inside your refrigerator while you're at the grocery store so you can see what you need to buy.

Small business mobile apps

It doesn't matter how long you've been in business or how small your company may be, you can probably benefit from mobile app development. The aim is to increase sales revenue.

Apps also improve the user experience and help businesses stay competitive in a market that's always changing. You can't afford to fall behind, so you need to stay up to date with the latest technology.

Increased mobile payments

As mobile security improves and global app revenue rises, we'll see an increase in mobile payments as well. Among them are:

- bank apps
- PayPal
- Venmo
- Google Pay
- Samsung Pay
- Apple Pay

Again, this relates to mobile security. There is a direct correlation between how comfortable people feel making mobile payments and the increase in mobile payment popularity.

Hybrid apps

Mobile app development can be expensive. Business owners have weighed the pros and cons of native and hybrid app development.

While both have their upsides, native development is more expensive. Furthermore, native apps can be built only for one platform at a time.

But hybrid apps give people the opportunity to launch their apps on the Google Play Store and Apple App Store simultaneously for a fraction of the cost.

That's another reason why more mobile apps are available for download, which contributes to the rising global app revenue as well.

Personal mobile devices in the workplace

Research indicates 87% of businesses depend on their employees to access work resources from their personal mobile devices. The concept is known as BYOD, or bring your own device.

Some companies disallowed BYOD and said it was due to security concerns. However, mobile security is improving, so now businesses are adapting and changing their policies.

Research shows there is a 34% increase in productivity when employees are allowed to use their personal mobile devices for work, which makes sense. They are used to handling these devices on a daily basis. It's easier for them to navigate and stay organized. This also helps businesses cut back on costs since they don't need to pay for new hardware.

Biometrics advancements

Biometrics are used to enhance security for mobile devices. Examples of biometrics include:

- voice recognition
- facial recognition
- signature recognition
- fingerprint recognition

Your current device may have some of these features installed. We'll see a lot more of this moving forward.

For example, let's say you've got an iPhone that was released in the last couple of years. You're used to the fingerprint feature to unlock the device.

But now the new iPhone X has facial recognition software.

Other apps are using biometrics as well. For example, you may be required to use your fingerprint to make a mobile payment through some platforms.

Mobile cloud

The mobile computing has met cloud computing which has centrally coordinated applications and can be shared and delivered to a number of devices which will continue to grow horizontally as well as vertically in its span of spread. Cloud service software is acknowledged to the rich APIs. Cloud has become a point of integration for data management technologies and applications including Internet of thing, social media mobile computing or any advanced analytics. The upcoming trend will make cloud transform the basic idea of computing i.e. to access data to "cloud/client computing ". It is going to facilitate the incorporation of mobile technologies. The utility stated in mobile computing is not only limited to geographic location and temporal usage but also to liquid computing in which the concerned user can switch between devices and save the state of their data, not only this they are also privileged to alter a same data from a different device.

Advanced network capacity

4G or the fourth generation has succeeded the third-generation networking and had made its place in the internet market. It highly supports mobile computing and has led an ease of faster

access through the Internet. This network is likely to be succeeded by the 5G which promises to set the market on a completely new position.

Software oriented applications, networking, storage, security and data holders are maturing with advancements. It is set that the computing has to step ahead of static models and enter into the dynamic range of changing demands of the digital business entities.

Internet of Things

The expansion of Internet of things (IOT) is the best example of shift in computing model. Dubbed 'wearables', the gadgets which includes smartwatches, glasses, vehicles, virtual personal assistants, wrist phones, all these devices are there to access data and virtually mandate it. Internet of things (IOT) provide customers to challenge their increasing need for interacting with organizations. At present the consumer market is leading in utilizing the mobile computing. Analytics will keep on growing by the Internet of things creating large sources of data. Every app will grow into analytic app to support an answer to the digital screen users. The analytics will not only grow but is likely to become more advanced and pervasive for a better impact.

Long Term Evolution (LTE)

LTE is a standard for wireless communication of high-speed data for mobile phones and data terminals. It is based on the GSM/EDGE and UMTS/HSPA network technologies, increasing the capacity and speed using new modulation techniques. It is related with the implementation of fourth Generation (4G) technology.

WiMAX

WiMAX (Worldwide Interoperability for Microwave Access) is a wireless communications standard designed to provide 30 to 40 megabit-per-second data rates, with the latest update providing up to 1 Gbit/s for fixed stations. It is a part of a fourth generation or 4G wireless-communication technology. WiMAX far surpasses the 30-metre wireless range of a conventional Wi-Fi Local Area Network (LAN), offering a metropolitan area network with a signal radius of about 50 km. WiMAX offers data transfer rates that can be superior to conventional cable-modem and DSL connections, however, the bandwidth must be shared among multiple users and thus yields lower speed in practice.

Near Field Communication

Near Field Communication (NFC) is a set of standards for smartphones and similar devices to establish radio communication with each other by touching them together or bringing them into close proximity, usually no more than a few centimeters. Present and anticipated applications include contactless transactions, data exchange, and simplified setup of more complex communications such as Wi-Fi. Communication is also possible between an NFC device and an unpowered NFC chip, called a "tag".

Global Positioning System (GPS)

The Global Positioning System (GPS) is a space-based satellite navigation system that provides location and time information in all weather, anywhere on or near the Earth, where there is an unobstructed line of sight to four or more GPS satellites. The GPS program provides critical capabilities to military, civil and commercial users around the world. In addition, GPS is the backbone for modernizing the global air traffic system, weather, and location services.

Larger Displays on Smartphones

Larger displays on smartphones have been in demand because smartphones are no longer meant only to make or receive calls. Owners of smartphones now want to view videos or have a better experience reading e-books, browsing websites or playing games. Phablets are becoming more commonplace, and increasing numbers of phones are being manufactured with large screens.

Non-Removable Battery and Memory

Another development in the smartphone space has been the introduction of non-removable batteries in many models. In addition to that, many smartphones are doing away with the mounted SD card option.

Non-removable batteries have been found to provide longer service life than removable batteries. Additionally, manufacturers are able to make phones slightly smaller by not needing to include access panels and/or mechanisms normally needed for removable batteries. However, for

smartphones that hang because of a software malfunction, users are no longer able to restart the phone by removing the battery.

Ingress Protection for Mobile Phones

Many companies are trying to develop glass for their smartphones that complies with the European Committee for Electrotechnical Standardization. Due to this, many smartphones are being given protection in the following ways:

- Protection from solid objects or materials
- Protection from liquids such as water
- Protection against mechanical impact

M-Commerce

Various analysts believe positive trend in mobile purchases will continue as more and more consumers adapt to m-commerce. Increasing popularity of Apple Pay and Google Wallet will facilitate purchases using the mobile phones instead of debit or credit cards. This will require developers to build a mobile application that can process transactions without the need of physical debit/credit cards or cash. Coupled with wearables that can process payments m-commerce will take a different shape. Beyond data collection and predictive analytics, wearables will also play a key role in the future of mobile payments and customer loyalty.

Conclusion

Technology is constantly evolving. Just compare your current smartphone to the phone you had five or ten years ago.

As we continue, we'll see several different trends. Older mobile technology will be improved, and newer technology will be introduced.

The idea is to understand how consumers react to these changes. They are the ones who shape these trends.

Identify the upward trends, and adapt accordingly to meet the needs of your mobile customers.